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Professor Ahmedouri
Editor-in-Chief
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Dear Professor Louri,

Please find enclosed the major revision of manuscript TSUSC-2026-07-0200, “Shielded Q-learning for Thermally-Safe, Energy-Aware Multi-Model Intrusion Detection Scheduling on Lightweight Edge Devices.”

We have revised the manuscript extensively in response to the Associate Editor and reviewers. The revision retains the submitted FSSQL-R method and operating constraints, but replaces the earlier replay-centered deployment evidence with new hardware experiments. The empirical implementation was rebuilt from the submitted method definitions and frozen before evaluation; the revised tables are replacement evidence rather than a claim of numerical reproduction of the earlier replay tables. The study now includes independent calibration on Raspberry Pi 3B+ and Raspberry Pi 4B 8GB, online closed-loop execution, a no-training Threshold controller, matched-seed statistics, Q-learning stability analysis, controller/switching overhead, direct POWER-Z KM003C DC input-energy measurement, and one-time fixed-threshold evaluation on the official UNSW-NB15 test set.

The new evidence led us to narrow several claims. We no longer present shielding as universally necessary, Q-learning as universally superior, or the guard as an unconditional complete-pipeline safety guarantee. We retain adverse results, including the stronger Safe-Greedy result on Pi 3B+, the strong Threshold baseline on Pi 4B, and the absence of a consistent physical-energy advantage for FSSQL-R. We also corrected the submitted protocol’s use, as described, of test-set detector statistics in the scheduler objective: the revised reward uses validation-partition TPR/FPR only, and the official test set remained sealed until all development and hardware decisions were frozen. Recent 2024–2025 TSUSC, TMC, and TIFS work has also been added to strengthen the positioning.

A detailed point-by-point response is provided as a separate Summary of Changes document, and supplemental material contains additional calibration, statistical, energy, and official-test details.

Thank you for your consideration of the revised manuscript.

Sincerely,

Nuonan Ouyang
Corresponding Author